BCSC009: SOFTWARE ENGINEERING

Objective: *Be employed in industry, government, or entrepreneurial endeavors to demonstrate professional advancement through significant technical achievements and expanded leadership responsibility.*

Credits:03 **L-T-P-J:3-0-0-0**

Module No.	Content	Teaching Hours
I	**Introductory Concepts**: The evolving role of software – characteristics, components and applications. **Process Models:** Waterfall Model, Prototyping, Incremental, Spiral, RAD. **Software Requirement Specification**: Requirement Process, SRS Components, Requirement Specifications with Use Cases Diagram, Requirements Validation. **Software Project Planning:** Project Planning Objectives. **Software Metrics:** Size, Function Point, Staffing, Project Estimation Methods– Decomposition Techniques; Empirical Estimation Models – COCOMO Model.	13
II	**Function-Oriented Design**: Problem Partitioning, Abstraction, Top Down and Bottom Up Design. **Module-Level Concepts:** Coupling, Cohesion, Design Notation and Specification - Structure Charts; Structured Design Methodology - Data Flow Diagram. **OO Analysis and OO Design:** OO Concepts, Introduction to UML Design Patterns. **Design Verifications:** Design Walkthroughs, Critical Design Review, Consistency Checkers. **Coding:** Coding Process, Verification - Code Inspections, Static Analysis, Proving Correctness; Metrics- Size Measures and Complexity Metrics.	13
III	**Testing Fundamentals**: Test Case Design, White Box Testing, Basis Path Testing, Control Structure Testing, Black Box Testing Strategies, Unit Testing, Integration Testing, Validation Testing, Reliability Estimation, Basic Concepts and Definitions, Reliability Model. Software Quality, ISO 9000 Certification for Software Industry, SEI Capability Maturity Model. **Software Configuration Management:** Introduction to SCM, Version Control and Change Management. **Risk Management:** Risk Mitigation, Monitoring, and Management. **Software Maintenance**: Models, Cost of Maintenance, Re-engineering, Reverse Engineering.	14

Text Books:

- R. S. Pressman (2010), "*Software Engineering: A Practitioners Approach*", 7thEdition, McGraw Hill.

Reference Books:

- K. K. Aggarwal and Yogesh Singh (2008), "*Software Engineering*", 3rd Edition, New Age International Publishers.
- Rajib Mall (2009), "*Fundamentals of Software Engineering*", 3rd Edition, PHI Publication.
- R.E Fairley (2004), "*Software Engineering*", McGraw Hill.
- Sommerville (2010), "*Software Engineering*", 9th Edition, Pearson Education.

Outcome:

Course Outcome

- CO1: Understand the basic concepts of software engineering.
- CO2: Apply software processes to solve real world problems.
- CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
- CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
- CO5: Understand the basic concepts of OO analysis and design.
- CO6: Develop the test cases to validate the software.
- CO7: Understand the basic models of software Quality and maintenance.

Printed Pages: **University Roll No. .....**

End Term Examination, Odd Semester 2022-23
B.Tech (CSE), II Year, III Semester
B.Tech (CCV, DA, IIOT, AIML, CSF), III Year, V Semester
Software Engineering (BCSC0009)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt AllQuestions 4 X 5 = 20 Marks

No.	Detaii of Question	Marks	CO	BL	KL
1	Give the parametric comparison of Waterfall, Spiral and Prototyping model.	4	1	An	C
2	Suppose you are working on Requirement Analysis & Specification phase of a project. you are in need to create a joint team of customers and developers just to understand the expectations and propose a set of requirements. Now identify the best requirement elicitation technique in this case & what guidelines will you follow.	4	2	An	F
3	What is Module Coupling? Explain any four type of Coupling with the help of an example.	4	5	U	C
4	Consider a program for determining the Previous Date. Its input is a triple of day, month and year with the values in the range January $\leq$ month $\leq$ December $1 \leq$ day ≤ 30 $1950 \leq$ year ≤ 2075 Design the Robustness test cases for the same.	4	6	E	P
5	Consider a program given below for the selection of the largest of numbers	4	4	E	P

```
main()
   {
     float A,B,C;
     printf("Enter three values\n");
     scanf("%f%f%f",  &A,&B,&c);
     printf("\n Largest value is");
     if (A>B)
          {
               if(A>C)
                    printf("%f\n",A);
               else
                    printf("%f\n",c);
          }
          else
          {
               if(C>B)
                    printf("%f\n",C);
               else
                    printf("%f\n",B);
          }
     }
```

Draw the flow graph and DD graph. Calculate the Cyclomatic Complexity by all the methods.

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	What is the difference between Reverse Engineering and Re-engineering? Explain with diagram.	3	7	U	F
7	A software project has development effort of 500 PM. It is assumed that 10% code will be modified per year. Some of the cost multipliers are given as: 1. Required software Reliability (RELY) : high= 1.15 2. Date base size (DATA) : high= 0.86 3. Analyst capability (ACAP) : high= 0.82 Other multipliers are nominal. Calculate the Annual Maintenance Effort (AME).	3	7	An	P

8	An authorized User can withdraw or deposit the money using an ATM machine. Create sequence diagram by showing all the necessary steps of ATM machine. Also discuss the purpose of creating sequence diagram.	3	4	An	F
9	Differentiate between • Alpha and Beta Testing • Functional Testing and Structural Testing • Validation and Verification	3	6	R	F
10	List the difference of CMM and ISO9001. Explain SEI-CMM Level.	3	7	R	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Why is maintenance programming becoming more challenging than new development? Explain the steps of software maintenance with help of a diagram. What is ripple effect? Enlist various categories of maintenance.	5	6	U	F
12	Consider the program for the selection of the largest of numbers in (Section-A Question-5). Compute the following- 1. Vocabulary of the program 2. program length 3. Program volume 4. Program level 5. potential volume	5	4	E	C
13	Consider a simple program to classify a triangle. Its inputs is a triple of positive integers (say x, y, z) and the date type for input parameters ensures that these will be integers greater than 0 and less than or equal to 100. The program output may be one of the following words: [Scalene; Isosceles; Equilateral; Not a triangle]. Identify the equivalence class test cases for output and input domain.	5	6	E	P

Printed Pages:04 **University Roll No: .....**

End Term Examination, Odd Semester 2021-22
B.Tech (CSE), IInd Year, IIIrd Semester

BCSC 0009: Software Engineering

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

(I) Explain Agile Software Development model with the help of diagram.

(II) Consider a Pizza Delivery system. Where client can order a pizza on pizza App. Pizza company collect all the orders, prepare the Pizza, deliver the Pizza, receive the payment from client and give payment to his employee's.
Draw DFD level-0, level-1 by considering above scenario.

(III) An authorized User can withdraw or deposit the money using an ATM machine. Create sequence diagram by showing all the necessary steps of ATM machine. Also discuss the purpose of creating sequence diagram.

(IV) What is the difference between Reverse Engineering and Re-engineering? Explain with diagram.

(V) Consider a program for the determination of the nature of roots of a quadratic equation. Its input is a triple of positive integers (say a,b,c) and values may be from interval [0,100]. The program output may have one of the following words.
[Not a quadratic equation; Real roots; Imaginary roots; Equal roots]
Design the Robust test cases.

(VI) What is the use of Structure chart in system design? Create Structure chart by considering the following code:

```
dim num1, num2 as integer

sub calculateAverage()
 dim avg as integer
```

```
 inputNums()
 avg = average(num1, num2)
 outputAvg(avg)
end sub

function average(a,b)
 return (a + b) / 2
end function

sub inputNums()
 num1 = console.readline()
 num2 = console.readline()
end sub

sub outputAvg(x)
 console.writeline("average = " & x)
end sub
```

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

(I) What is Module Cohesion? Explain any five Cohesion types.

(II) Consider a program to find Largest number among three numbers. Design the test cases using Decision Table Based Testing.

(III) Design a flow chart for Test Driven Development Coding process.

(IV) Explain the following:
(a) Modularization Curve (b) Module Coupling (c) Unit Testing

(V) Let us consider an example of software application. There is function of software application that accepts only particular number of digits, not even greater or less than that particular number.
Consider an OTP number that contains only 6 digit number, greater and even less than six digits will not be accepted, and the application will redirect customer or user to error page. If password entered by user is less or more than six characters, that equivalence partitioning method will show an invalid OTP. If password entered is exactly six characters, then equivalence partitioning method will show valid OTP.
Design the classes and test cases using equivalence class testing technique.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

(I) Explain SEI- CMM Model of software quality. Define any five important parameters to judge the quality of a software.

(II) Explain Taute Maintenance Model with the help of diagram.
A software project has development effort of 500 PM. 90 KLOC is added and 60 KLOC is deleted from of software due to maintenance work per year. Some of the cost multipliers are given as:
1. Required software Reliability (RELY) : high= 1.15
2. Date base size (DATA) : high= 0.86
3. Analyst capability (ACAP) : high= 0.82
Other multipliers are nominal. Calculate the Annual Maintenance Effort (AME). size of the project is 600KLOC.

(III) Consider the program(given in below Question IV). List out the operators and operands and also calculate the values of software science measures like η, N, V.

(IV) Draw the Control flow graph for the given program. Find out all independent paths and Design the test cases that will guarantee that all statements in the program have been tested. Also verify your answer using cyclomatic complexity.

```
#include <stdio.h>
void main()
{
  int marks;
  printf("Enter your marks?");
  scanf("%d",&marks);
  if(marks > 85 && marks <= 100)
  {
    printf("Congrats ! you scored grade A ...");
  }
  else if (marks > 60 && marks <= 85)
  {
    printf("You scored grade B + ...");
  }
  else if (marks > 40 && marks <= 60)
  {
    printf("You scored grade B ...");
```

```
    }
    else if (marks > 30 && marks <= 40)
    {
       printf("You scored grade C ...");
    }
    else
    {
       printf("Sorry you are fail ...");
    }
}
```

Printed Pages :4 University Roll No: .....

End Term Examination, Odd Semester 2019-20

Program: B.Tech(CS), Year: II Semester: III

Subject Code:BCSC0069 Subject Title: Software engineering

Time: 3 Hour Maximum Marks: 50

Section-A

Note: Attempt all questions. 7 x 5 = 35 Marks

I. Discuss the elicitation technique which promotes the creative thinking and new ideas. Consider the following activities of Restaurant System:

- Customer Order
- Serve Product
- Collect Payment
- Produce Product
- Store Product
- Order Raw Materials
- Pay for Raw Materials
- Pay for Labor

Consider the above scenario and create the following:

a) Use Case Diagram

b) DFD level-0, level-1.

OR

Assume that you are the technical manager of a software development organization. A client approached you for a software solution. The problems stated by the client have uncertainties which lead to loss if it not planned and solved. Which software development model you will suggest for this project-justify. Explain the model with its pros and cons and neat sketch.

II. Compare the Walston-Felix model with the SEL model on a software development expected to involve 8 person-years of effort.

a) Calculate the number of lines of source code that can be produced.

b) Calculate the duration of the development.

c) Calculate the productivity in LOC/PY

d) Calculate the average manning

III. Differentiate between structure chart and flow chart. Consider a program that reads a number from array, sort the number using any sorting algorithm and print the sorted list of elements. Draw structure chart for this problem and clearly show the data and control flow.

OR

Discuss the various software planning objectives. Also enlist various function oriented design notations. Design the structured chart for the following code with proper leveling:

```
subhowManyThrees()
dimnum1,count,totalasinteger
num1=startMsg()
count=0
total=0
whilenum1>0do
        checkNumber(count,total,num1)
num1=num1−1
endwhile
endMsg(count)
endsub
subcheckNumber(byRefc,byReft,byValn)
        IfnMOD3=0Then
                c=divBy3(c)
        Else
                t=add(n,t)
        EndIf
endsub
functiondivBy3(x)
        returnx+1
endfunction
functionadd(n,t)
        returnn+t
endfunction
functionstartMsg()
        console.writeline("program started, enter your number")
        returnconsole.readline()
endfunction
subendMsg(n)
        console.writeline("number of threes : "&n)
endsub
```

IV. What do you understand by structure Design methodology? Apply all four steps of SDM for given scenario:
GLA University has to design a software system for Result Management system.(make necessary assumptions)

V. Consider the fallowing program and Draw flow graph, DD path graph, identify independent paths; Write the test cases, Calculate cyclomatic complexity by all four methods.

```
#include <stdio.h>
void main ()
{
int number1, number2;
printf("Enter two integers: ");
scanf("%d %d", &number1, &numt er2);
if (number1 >= number2)
  {
if (number1 == number2)
   {
printf("Result: %d = %d",number1,number2);
   }
else
   {
printf("Result: %d > %d", number1, number2);
   }
  }
else {
printf("Result: %d < %d",number1, number2);
  }
getch();
}
```

VI. Admission to a professional course is subject to the following Conditions:

1. Marks in Mathematics >= 60
2. Marks in Physics >= 50
3. Marks in Chemistry >= 40
4. Total in all three subjects >= 200 or Total in Mathematics and Physics >= 150

If aggregate marks of an eligible candidate are more than 225, he/she will be eligible for honors course; otherwise he/she will be eligible for pass

course.

The program reads the marks in the three subjects and generates the following outputs:

(a) Not Eligible

(b) Eligible to Pass Course

(c) Eligible to Honors Course

Design test cases using decision table testing technique and Equivalence class testing technique.

VII. Draw the framework of Automation testing. Why Automation Testing is needed? Explain Automated Testing process of Selenium with the help of examples (By considering different test cases).

Section-B

I. ***Note: Attempt all questions.*** **3 x 2 =6 Marks**

(a) Differentiate between Reverse and Re Engineering. In which scenario reverse engineering is required. Give the illustration that shows the application of reverse engineering at different level of abstraction.

(b) The development effort for a software project is 500 person months. The empirically determined constant (K) is 0.3. The complexity of the code is quite high and is equal to 8. Calculate the total effort expended (M) if

(i) Maintenance team has good understanding of the project (d=0.9)

(ii) Maintenance team has poor understanding of the project (d=0.1)

(c) Explain with the help of example why single entry, single exit modules make testing easier during maintenance.

II. ***Note: Attempt all questions.*** **3 x 3 =9 Marks**

(a) Consider a simple program to classify a triangle. Its inputs is a triple of positive integers (say x, y, z) and the date type for input parameters ensures that these will be integers greater than 0 and less than or equal to 100. The program output may be one of the following words:

[Scalene; Isosceles; Equilateral; Not a triangle]

Define all the input and output domain Equivalence Classes.

(b) Discuss the relationship between quality factors and quality criteria in McCall's software quality model.

(c) Discuss how the CMM encourages continuous improvement of the software process. Also enlist various key process areas of CMM at various maturity levels.

University Roll no.....

End Term Examination, Even semester 2018-19

B.Tech. II Year/ IV Semester

Software Engineering (BCSC 0009)

Uni. Roll No: .....

Time: 3 Hrs. Total Marks: 50

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions (7x5=35)

I. Consider a mail order company that distributes CDs and tapes at discount prices to its members. When an order processing clerk receives an order form, she verifies that the sender is a member by checking the MEMBER FILE.
If the sender is not a member, the clerk returns the order along with a membership application form. If the customer is a member, the clerk verifies the order item data by checking the ITEM FILE.
Then the clerk enters the order data and saves it to the DAILY ORDERS FILE. At the same time the clerk also prints an invoice and shipping list for each order, which are forwarded to the ORDER FULFILLMENT DEPARTMENT for processing there.
Consider the above scenario and create DFD level-0, level-1, level-2.

II. Consider a C project with the following parameters.
(i) External Inputs:
(a) 10 with low complexity
(b) 15 with average complexity
(c) 17 with high complexity
(ii) External Outputs:
(a) 6 with low complexity
(b) 13 with high complexity
(iii) External Inquiries:
(a) 3 with low complexity
(b) 4 with average complexity
(c) 2 high complexity

(iv) Internal logical files:
(a) 2 with average complexity
(b) 1 with high complexity
(v) External Interface files:
(a) 9 with low complexity
In addition to above, system requires
i. Significant data communication
ii. Performance is very critical
iii.Designed code may be moderately reusable
iv. System is not designed for multiple installation in different organizations.
Other complexity adjustment factors are treated as average. Calculate the Average staff size and Productivity of the project using COCOMO Model.

III. Define the role of structure Design methodology in designing a software. Explain all four steps of SDM for given scenario:
Company have to design a software system to determine the number of unique words in the file given by user.

IV. Compute vocabulary, program length, volume of the given code:

```
void arrange_data (int a, int n)
{
int i, j, t;
      if( n < 2 )
      return;
 for ( i=0 ; i <n-1; i++ )
  {
  for ( j=i+1 ; j < n ; j++ )
     {
       if( a[i] > a[j] )
          {
            t= a[i]; a[i] = a[j]; a[j] = t;
          }
     }
 }
}
```

V. Consider the fallowing program and Draw flow graph, DD path graph, identify independent paths, Write the test cases, Calculate cyclomatic complexity by all four methods.

```
#include <stdio.h>
Void main ()
{
  int a = 100;
  if( a == 10 )
{
     printf("Value of a is 10\n" );
  }
else if( a == 20 )
 {
    printf("Value of a is 20\n" );
  }
 else if( a == 30 )
 {
    printf("Value of a is 30\n" );
  }
 Else
 {
     printf("None of the values is matching\n" );
  }
  printf("Exact value of a is: %d\n", a );
  getch();
}
```

VI. Expiain Equivalence Class Testing Method.

Consider the following scenarios and write the test cases according to Equivalence Class Testing Method for each scenarios.

Assume, we have to test a field which accepts Age 18 – 56 for man and Age 20-52 for women.

Assume, we have to test a filed which accepts a Mobile Number of ten digits.

Suppose you have very important tool at office, accepts valid User Name and Password field to work on that tool, and accepts minimum 8 characters and maximum 12 characters. Valid range 8-12, Invalid range 7 or less than 7 and Invalid range 13 or more than 13.

VII. Define Categories of Software Maintenance. Which category of software maintenance consumes maximum cost of maintenance and why? Explain Taute Maintenance Model with the help of diagram.

Section B

Note: Attempt All Question. (3x2=6)

Part-1

I. Differentiate between Alpha and Beta testing

II. What is the role of Driver and stub in Unit Testing? Explain with diagram.

III. What is the need of Reverse Engineering? Explain with diagram.

Part-2 (3x3=9)

I. Explain Mutation Testing. Consider a project of size 500KLOC needed maintenance, under maintenance company is going to delete 30KLOC and add 70KLOC in project. Compute an estimate for annul maintenance effort.

II. Explain CMM model and enlist the KPA of level 3 and level 5.

III. Explain ISO-9126 software quality model with the help of diagram.